testbench运行指南

1.下载iverilog

网址如下：http://bleyer.org/icarus/ windows下载以下版本即可

iverilog-v12-20220611-x64_setup [18.2MB]

运行安装程序，能打勾的地方全部打勾，直到安装完成。

打开cmd，输入**iverilog**进行测试，出现如下场景即为安装成功。

```
C:\Users\Administrator>iverilog
iverilog: no source files.

Usage: iverilog [-EiSuvV] [-B base] [-c cmdfile|-f cmdfile]
                [-g1995|-g2001|-g2005|-g2005-sv|-g2009|-g2012] [-g<feature>]
                [-D macro[=defn]] [-I includedir] [-L moduledir]
                [-M [mode=]depfile] [-m module]
                [-N file] [-o filename] [-p flag=value]
                [-s topmodule] [-t target] [-T min|typ|max]
                [-W class] [-y dir] [-Y suf] [-l file] source_file(s)

See the man page for details.

C:\Users\Administrator>
```

编译并运行

将压缩包中的alu_tb_1.v和alu_tb_2.v，以及你实现的alu.v文件放于同一目录下，进行编译并运行。

编译

输入如下命令行进行编译

```
iverilog -o alu_test alu.v alu_tb_1.v
```

其中alu_test是指定生成的编译文件。预期结果如下：

```
:\Users\Administrator\Desktop>iverilog -o alu_test alu.v alu_tb_1.v

:\Users\Administrator\Desktop>
```

运行

输入如下命令行进行测试

```
vvp alu_test
```

两个测试文件的预期结果如下所示：

```
C:\Users\Administrator\Desktop>iverilog -o alu_test alu.v alu_tb_1.v

C:\Users\Administrator\Desktop>vvp alu_test
=================== ALU 详细测试报告 ===================

测试1:                     10 + 20 = 30 -> ? 通过
   输入: A=0a, B=14, Op=1000
   输出: 期望=1e, 实际=1e

测试2:                     30 - 15 = 15 -> ? 通过
   输入: A=1e, B=0f, Op=1001
   输出: 期望=0f, 实际=0f

测试3:                     13 << 2 = 52 -> ? 通过
   输入: A=02, B=0d, Op=0000
   输出: 期望=34, 实际=34

测试4:                    208 >> 2 = 52 -> ? 通过
   输入: A=02, B=d0, Op=0010
   输出: 期望=34, 实际=34

测试5:                  -48 >>> 2 = -12 -> ? 通过
   输入: A=02, B=d0, Op=0011
   输出: 期望=f4, 实际=f4

测试6:               0xAA & 0xCC = 0x88 -> ? 通过
   输入: A=aa, B=cc, Op=1100
   输出: 期望=88, 实际=88

测试7:               0xAA | 0xCC = 0xEE -> ? 通过
   输入: A=aa, B=cc, Op=1101
   输出: 期望=ee, 实际=ee

测试8:               0xAA ^ 0xCC = 0x66 -> ? 通过
   输入: A=aa, B=cc, Op=1110
   输出: 期望=66, 实际=66

测试9:            ~(0xAA | 0xCC) = 0x11 -> ? 通过
   输入: A=aa, B=cc, Op=1111
   输出: 期望=11, 实际=11

=================== 测试总结 ===================
?? 通过率: 9/9 (100.0%)
?? 恭喜! 所有测试用例都通过了!
?? 你的ALU设计完全正确!
alu_tb_1.v:74: $stop called at 135000 (1ps)
** VVP Stop(0) **
** Flushing output streams.
** Current simulation time is 135000 ticks.
>
```

```
===================================================
[PASS] Subtraction: 50 - 30 = 20
      Expected: 14 (20), Got: 14 (20)
[PASS] Subtraction Negative: 10 - 25 = -15
      Expected: e1 (225), Got: f1 (241)
[PASS] Subtraction Equal: 100 - 100 = 0
      Expected: 00 (0), Got: 00 (0)

===================================================
SHIFT TESTS - LOGICAL LEFT SHIFT (Op=0000)
===================================================
[PASS] Left Shift 1: 10 << 1 = 20
      Expected: 14 (20), Got: 14 (20)
[PASS] Left Shift 3: 1 << 3 = 8
      Expected: 08 (8), Got: 08 (8)
[PASS] Left Shift Overflow: 16 << 4 = 0
      Expected: 00 (0), Got: 00 (0)

===================================================
SHIFT TESTS - LOGICAL RIGHT SHIFT (Op=0010)
===================================================
[PASS] Logical Right Shift 1: 32 >> 1 = 16
      Expected: 10 (16), Got: 10 (16)
[PASS] Logical Right Shift 2: 240 >> 2 = 60
      Expected: 3c (60), Got: 3c (60)
[PASS] Logical Right Shift 4: 128 >> 4 = 8
      Expected: 08 (8), Got: 08 (8)

===================================================
LOGIC TESTS - BITWISE AND (Op=1100)
===================================================
[PASS] Bitwise AND: F0 & CC = C0
      Expected: c0 (192), Got: c0 (192)
[PASS] Bitwise AND Mask: 0F & DA = 0A
      Expected: 0a (10), Got: 0a (10)
[PASS] Bitwise AND Clear: 00 & FF = 00
      Expected: 00 (0), Got: 00 (0)

===================================================
LOGIC TESTS - BITWISE OR (Op=1101)
===================================================
[PASS] Bitwise OR: AA | 55 = FF
      Expected: ff (255), Got: ff (255)
[PASS] Bitwise OR Set: 0F | A0 = AF
      Expected: af (175), Got: af (175)
[PASS] Bitwise OR Identity: FF | 00 = FF
      Expected: ff (255), Got: ff (255)

===================================================
FINAL TEST RESULTS
===================================================
Total Tests: 18
Passed Tests: 18
Failed Tests: 0
Score: 100.0%
PERFECT! All tests passed!
===================================================
Simulation completed at time 300000
alu_tb_2.v:391: $stop called at 300000 (1ps)
** VVP Stop(0) **
** Flushing output streams.
** Current simulation time is 300000 ticks.
>
```

注意

第一个测试文件通过即可，第二个测试文件为附加测试。如果遇到iverilog配置问题或其他原因导致测试文件不可用，可用波形文件图代替，评分将酌情降低。